MILITARY SPECIFICATION

BOTTLE ASSEMBLIES, VACUUM, AIRCRAFT

1. SCOPE

1.1 Scope. This specification covers the general requirements for metal vacuum bottle assemblies for the purpose of storing and dispensing hot or cold liquids in aircraft. The assemblies are designed for use while installed either in the horizontal or vertical (inverted or up-right) position without removal of the bottle from the installation bracket. In addition, liquids may be dispensed via a single delivery tube for crew members without the pressure helmet and also may be dispensed by a combination of delivery and vent tube to crew members wearing the pressure helmet.

1.2 Classification.

1.2.1 Type and Sizes.

Type I - Horizontal installation (Figure 1).
Type II - Vertical inverted installation (Figure 2).
Type III - Vertical upright installation (Figure 3).

Size - 1 = 1 Quart
Size - 2 = 2 Quart

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-403	Plastic Molding Material, Polytetrafluroethylene (TFE - Fluorocarbon)
QQ-B-601	Brass, Castings
QQ-B-626	Brass Leaded and Non-Leaded: Rods, Shapes, Forgings and Flat Products with Finished Edges (Bars, Flat Wire and Strips)
QQ-B-613	Brass, Leaded and Non-Leaded, Plate, Rolled, Bar, Sheet and Strip

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QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickle Plating (Electrodeposited)
QQ-S-698	Steel, Sheet and Strip, Low Carbon
QQ-S-763	Steel Bars, Shapes, and Forgings-Corrosion Resisting
QQ-S-766	Steel Plates, Sheets and Strip-Corrosion Resisting
TT-E-489	Enamel, Alkyd, Gloss (For Exterior and Interior Surfaces)
TT-P-636	Primer Coating, Alkyd, Wood and Ferrous Metal
PPP-B-636	Box-Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-T-8606	Tubing, Steel, Corrosion-Resistant (18-8) Stabilized
MIL-M-20693	Molding Plastic, Polyamide (Nylon) Rigid
MIL-T-25458	Tubing, Rubber, Silicone, for In-Flight Feeding

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies)
FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), before production is commenced, a sample of the required vacuum bottle assembly covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for examination and approval in accordance with 4.2. Approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of the responsibility for compliance with the detailed provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

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3.2 Materials. Materials shall conform to the applicable specifications and requirements specified herein. Selection of materials not specifically defined shall be of the quality normally used in the manufacturer's standard commercial item.

3.2.1 Steel, carbon sheet. Carbon steel sheet shall conform to cold-rolled commercial quality of QQ-S-698.

3.2.2 Steel, corrosion-resisting. Corrosion resisting steel shall be in accordance with Classes 302, 303, 304, 305, Condition A of QQ-S-763 or QQ-S-766.

3.2.3 Brass. Brass sheet required for forming or drawing shall conform to QQ-B-613 or QQ-B-626. Brass required for machining shall conform to QQ-B-613 or QQ-B-601.

3.2.4 Plating. Where plating is specified, it shall be nickle and chromium in accordance with QQ-N-290 and QQ-C-320.

3.2.5 Plastics. Where plastics are used, they shall conform to L-P-403 and MIL-M-20693. The color shall be black unless otherwise specified.

3.2.6 Rubber tubing. The tubing used shall conform to MIL-T-25458 or equal.

3.2.7 Primer. Primer shall conform to TT-P-636.

3.2.8 Enamel. Enamel shall conform to Class B of TT-E-489. Color shall be lusterless black shade no. 37038 of FED-STD-595.

3.3 Design. The vacuum bottle assembly shall conform to Figure 1, 2, or 3 for whichever type is specified (see 6.2). The entire assembly shall provide for maximum sanitation and ease of cleaning.

3.4 Construction. Details not specifically defined herein shall be in accordance with the manufacturer's standard practice.

3.4.1 Vacuum bottle. The vacuum bottle shall consist basically of an inner shell, an outer shell, a bottom cap, a top cap with a spigot assembly, a drinking tube, a hand valve, and a vent tube when required. The inner and outer shells shall be joined by the manufacturer's standard commerical technique for welding or brazing. The space between the shells shall be evacuated of air. The inner shell shall be constructed of corrosion-resisting steel conforming to 3.2.2 and the outer shall shall be fabricated of cold-rolled steel conforming to 3.2.1.

3.4.2 Bottom cap. The bottom cap shall be fabricated of carbon steel conforming to 3.2.1.

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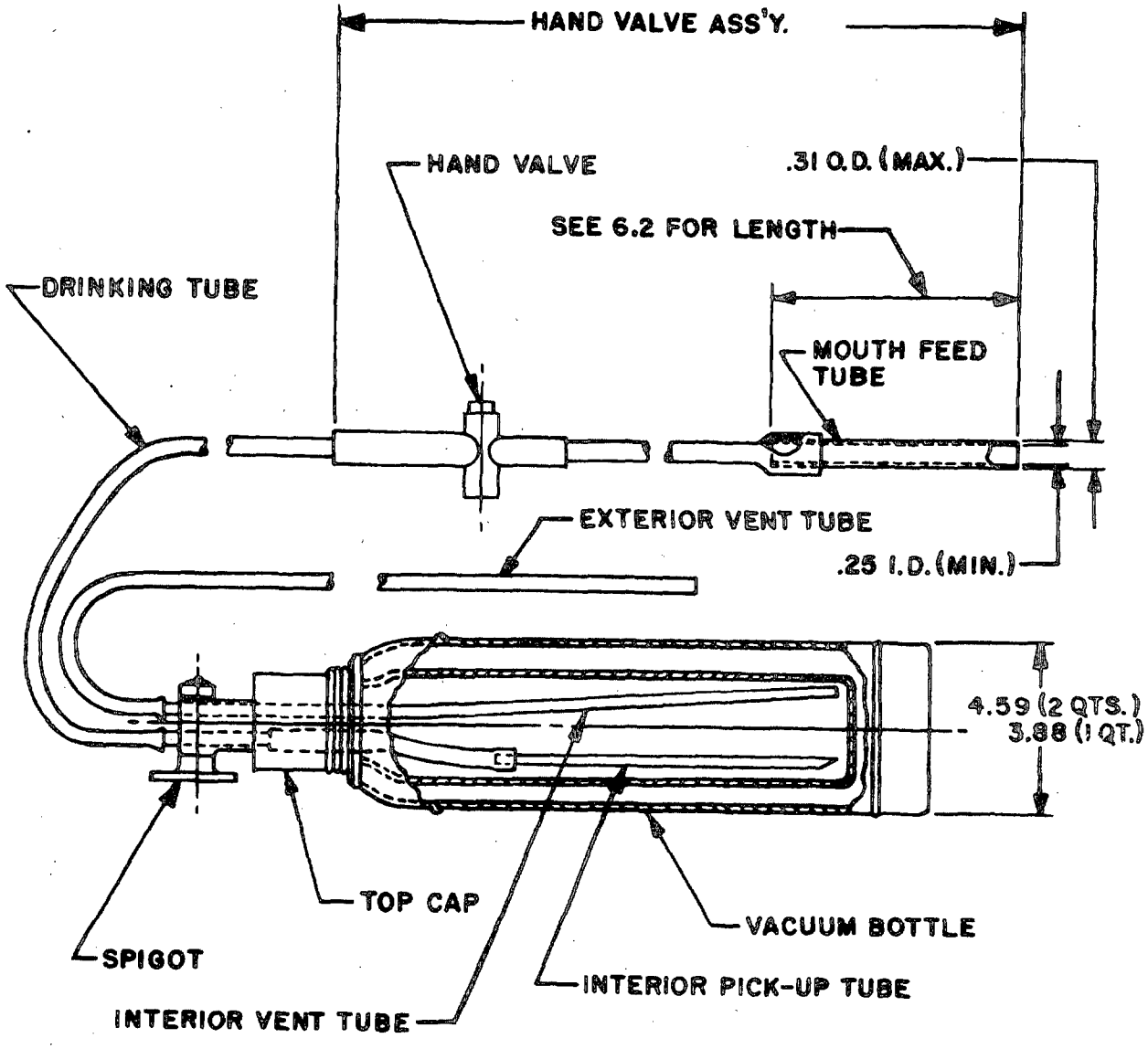


FIG. 1

HORIZONTAL INSTALLATION

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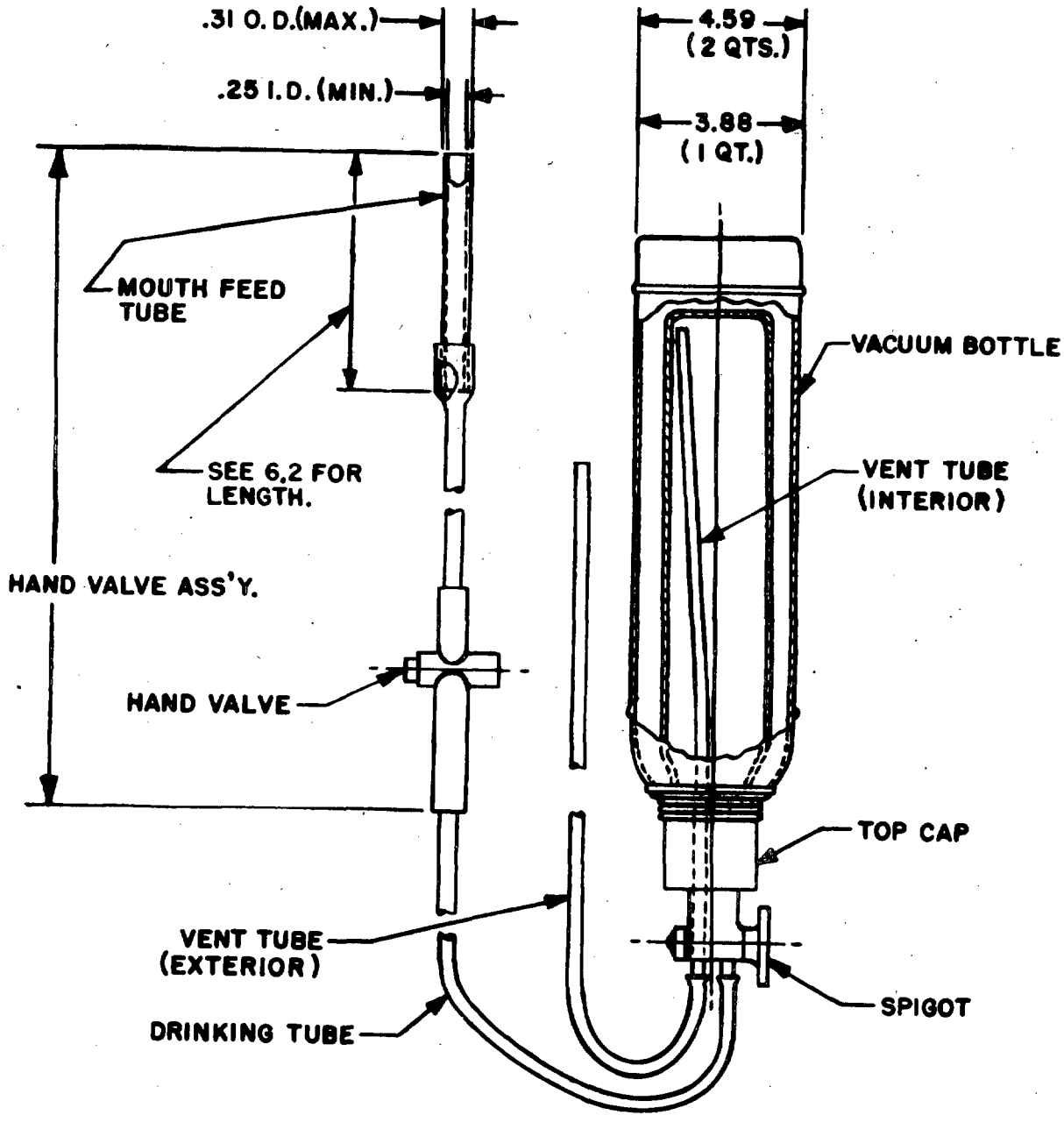


FIG. 2

VERTICAL INVERTED INSTALLATION

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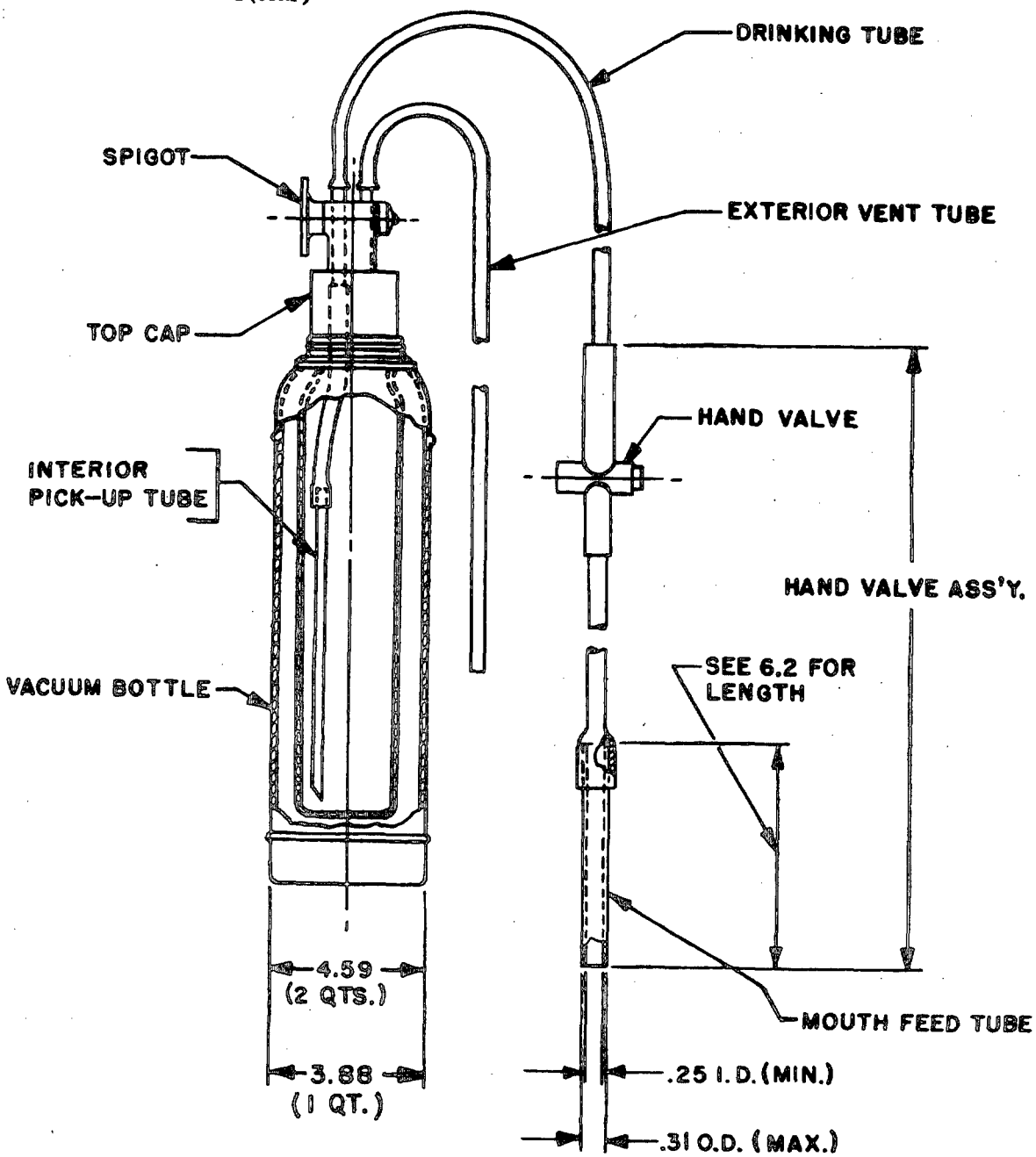


FIG. 3
VERTICAL UPRIGHT INSTALLATION

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3.4.3 Top cap. The vacuum bottle shall be provided with a screw-on top cap. The cap shall be designed with an integral spigot similar to that shown on Figures 1, 2, or 3. The cap and spigot shall be constructed from any of the following material options or a combination of them.

a. Brass conforming to 3.2.3.

b. Corrosion-resisting steel conforming to 3.2.2.

c. Plastic conforming to 3.2.5.

3.4.3.1 Spigot. The spigot shall be designed with a valve to open and close the liquid supply exit and an air vent simultaneously. The valve handle shall be parallel to the vacuum bottle when in the open position. The valve shall operate smoothly opening and closing without using lubricants.

3.4.4 Interior vent tube. An interior vent tube shall be installed in Type I and II vacuum bottles as shown on Figures 1 and 2. Type III vacuum bottles do not require an interior vent tube. The vent tubes shall be designed to bleed air into the vacuum bottle as liquid is drawn out. The tubes shall be non-toxic and comply to the requirements of 3.6.

3.4.5 Interior pickup tube. An interior pickup tube shall be installed in Type I and III vacuum bottles as shown on Figures 1 and 3. Type II vacuum bottles do not require interior pickup tubes. The interior pickup tube shall be capable of removing liquid from the lowest portion of the vacuum bottle. The tube shall be made partly of flexible tubing conforming to 3.2.6 and 3.6 and partly of stainless steel conforming to MIL-T-8606.

3.4.6 Drinking tube. Each vacuum bottle shall be equipped with a flexible drinking tube that is attached securely on one end to the liquid exit port of the spigot and on the other end to a hand valve assembly. The tube shall have an inside diameter of 1/4" and the length shall be specified by the procuring agency (see 6.2). The tube shall conform to 3.2.6 and 3.6.

3.4.6.1 Hand valve assembly. Each drinking tube shall be equipped with a hand valve assembly consisting of a hand valve, and a rigid mouth feed tube 5/16" maximum outside diameter and 1/4" minimum inside diameter sililar to that shown on Figures 1, 2, and 3. The length of the mouth feed tube shall be specified by the procuring activity to be 4" or 7" (see 6.2). The mouth feed tube material shall be nylaflow tubing type "H" or equal conforming to 3.6. The hand valve shall be made from any of the following material options:

a. Brass conforming to 3.2.3.

b. Corrosion-resisting steel conforming to 3.2.2.

c. Plastic conforming to 3.2.5.

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3.4.7 Exterior vent tube. Type I and II (and III where used with a pressure helmet) vacuum bottles shall be equipped with an exterior flexible vent tube conforming to 3.2.6 and 3.6. The tube shall have a minimum inside diameter of 1/8" and the length shall be specified by the procuring agency.

3.5 Finish. The surface finish shall be uniform throughout, free of irregularities, scratches and blemishes. Painted surfaces shall be completely coated and free of blisters, scratches, runs, sags, foreign matter, discoloration, and finger marks.

3.5.1 Finish, outer shell. The interior and the exterior surfaces of the outer shell shall be coated with a primer conforming to 3.2.7. Then both surfaces shall be painted with two coats of enamel conforming to 3.2.8.

3.5.2 Finish, inner shell. The interior surface of the inner shell shall have a uniform commercial No. 2B finish or better.

3.5.3 Finish, bottom cap. The bottom cap shall be plated on the exterior conforming to 3.2.4 and coated with primer on the interior per 3.2.7 and two coats of enamel per 3.2.8.

3.5.4 Finish, top cap. If the top cap is constructed of brass, then it shall be plated per 3.2.4. If the top cap is constructed of stainless steel, then it shall have a commercial 2B finish. If plastic is used, it shall be chromium plated on the exterior by the manufacturer's standard process and finished smoothly on the interior surface. The plastic interior shall be light green color.

3.6 Toxicity. The tubing, hand valve and spigot material shall bear the seal of approval of the National Sanitation Foundation Testing Laboratory or have been evaluated and found acceptable by another impartial agency using equivalent testing procedures and standards. The manufacturer shall cite the test report from any laboratory other than the NSF which may have conducted the tests.

3.7 Marking. Vacuum bottles shall be marked on the bottom in accordance with MIL-STD-130.

3.8 Workmanship. All joints in the vacuum bottle inner and outer shells shall be free of any defects which may adversely affect serviceability, appearance, and cleaning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Preproduction sample inspection. When a preproduction sample is required, it shall be examined for defects in Table II and subjected to the applicable tests in 4.5.2, 4.5.3, 4.5.4, 4.5.5, 4.5.6, 4.5.7 and 4.5.8.

4.3 Sampling for inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise specified herein.

4.3.1 Component and material inspection. In accordance with 4.1, components and material shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 Inprocess examination. Examination shall be made during the manufacturing process for the requirements in Table I, to establish that no deviation is made from indicated requirements. When nonconformance is noted, correction shall be made to affected items and process.

TABLE I INPROCESS EXAMINATION

	OPERATION	REQUIREMENT PARAGRAPH
1.	Outer shells primed internally and externally	3.5.1
2.	Outer shells coated internally and externally	3.5.1
3.	Inner shells have No. 2B finish or better	3.5.2
4.	Bottom caps plated externally, primed and coated internally	3.5.3
5.	Top caps: If brass, plated If CRES, No. 2B finish or better If plastic, smooth light green color	3.5.4

4.3.3 End item inspection. The lot shall consist of all vacuum bottle assemblies of the same type and size offered for inspection at one time.

4.3.3.1 Visual examination. The end items shall be examined for the applicable defects classified in Table II. The inspection level shall be II. The acceptable Quality Level (AQL) shall be 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

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TABLE II

EXAMINE	DEFECT	CLASSIFICATION MAJOR	MINOR
Finish (Exterior & interior shell)	Roughness or discoloration		X
	Rust, corrosion, or paint damaged	X	
	Color not lusterless black or as specified	X	
	Handle is not parallel with the bottle when the bottle's spigot valve is in the open position		X
Cap plate	Does not hold spigot firmly in place	X	
Any Gaskets & "O" Rings used	Imparts objectionable taste to the contents	X	
Vent tube for horizontal bottle	Does not extend into the inner shell properly (see Figure 1)		X
	Not bent to the opposite side from the handle installation		X
Pickup tube	Not assembled to the spigot properly		X
Flexible feed tube	Not attached to the liquid exit port of the spigot assembly	X	
	Hand valve not attached to the end of the flexible feed tube opposite from the liquid exit port	X	
Exterior vent tube	Not provided (when specified)		X
Welding	All welded surfaces not smooth or free from depressions or protrusions	X	
Dimensions	Not in accordance with Figures 1, 2, or 3, or paragraphs 3.4.6, 3.4.6.1, and 3.4.7.	X	

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4.3.3.2 Testing of the end item. Testing shall be performed on the samples selected from each inspection lot of bottles in accordance with Table III below for the characteristics specified therein. The inspection levels applicable to each inspection lot, shall be as indicated in table below.

All characteristics are applicable to individual unit requirements.

TABLE III

CHARACTERISTIC	TEST REQUIREMENT PARAGRAPH	INSPECTION LEVEL	ACC No or AQL
1. Assembly leakage	4.5.2	II	1.5
2. Bottle leakage	4.5.3	II	1.5
3. Functional test	4.5.4	S-2	2.5
4. Heat retention test	4.5.5	S-2	2.5
5. Spigot test	4.5.6	S-1	4.0
6. Hand valve test	4.5.7	S-1	4.0
7. Capacity	4.5.8	S-2	2.5

4.3.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and marking as required by section 5 are complied with. Defects shall be as indicated in Table IV. The sample unit shall be one shipping container fully prepared for delivery, except that it need not be sealed. The lot shall be all containers offered for inspection at one time. The inspection level shall be S-2. The AQL shall be 4.0, expressed in terms of defects per hundred units. Defects of closure shall be examined on shipping containers fully prepared for delivery and in accordance with the appendix of the container specification.

TABLE IV - VISUAL DEFECTS, PREPARATION FOR DELIVERY

EXAMINE	DEFECT
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Component missing or damaged
Workmanship	Inadequate application of cushioning material, tape banding, steel strapping or box flaps not secured
Contents	Weight of content of shipping containers exceeds requirements.

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4.5 Tests.

4.5.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 24 to 30 inches of mercury and at a temperature of 70 degrees plus or minus 10 degrees Fahrenheit.

4.5.2 Assembly leakage. The assembly, without the exterior vent and drinking tubes, shall be tested by attaching an air line to the delivery tube of the spigot with the vent tube closed off. The interior of the bottle shall be filled with an air pressure of not less than 15 pounds per square inch. The container shall then be immersed in water, while maintaining the above pressure, for not less than 5 minutes. Any leakage, as indicated by air bubbles from the container, shall be cause for rejection.

4.5.3 Bottle leakage. The container shall be tested after all assembly work has been accomplished, but before evacuation. An air line shall be attached to the bottom plug, and the inside shell shall be filled with water. An air pressure of not less than 50 nor more than 60 pounds per square inch shall be applied to the air line, and the interior of the bottle observed for air bubbles. The container, while still under pressure, shall then be immersed in water and the outside shell observed for air bubbles. Any evidence of leakage, as indicated by the formation of air bubbles, shall be cause for rejection.

4.5.4 Functional test. Each sample bottle assembly shall be filled with water and the cap and spigot put in place. Assemblies designed for vertical installation shall be mounted in the inverted position and the handle placed in the open position. There shall be a smooth steady flow of water until the bottle has been completely emptied. Assemblies designed for horizontal installation shall be mounted in the horizontal position and the handle placed in the open position. With the hand valve lower than the mouth of the bottle it shall be possible to easily empty the bottle except for a small residue of liquid.

4.5.5 Heat retention. The sample bottles shall be preheated with water at a temperature of plus 205 degrees Fahrenheit (plus or minus 2 degrees) for a period of 15 minutes. The bottles shall be emptied and refilled to their rated capacity with water at a temperature of 205 degrees Fahrenheit (plus or minus 2 degrees) and the cap and spigot assembly put in place and allowed to stand for a period of 6 hours. The temperature of the water after this period shall be not less than 150 degrees Fahrenheit for the 1 quart capacity and 160 degrees Fahrenheit for the 2 quart capacity.

4.5.6 Spigot test. The spigot shall be operated from the fully open to the fully closed position of the valve for a minimum of 500 times. The valve shall operate freely with no evidence of binding, and there shall be no evidence of breakdown of the handle.

4.5.7 Hand valve test. The hand valve shall be operated for a minimum of 500 times. The mechanism shall operate freely with no evidence of binding or breakdown.

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4.5.8 Capacity test. Samples shall be filled to the brim with water and poured into a measuring beaker and the volume shall be the specified capacity ± 1 fluid ounce.

4.5.9 Toxicity data. The supplier shall furnish the toxicological data required by 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2)

5.1.1 Level A. (See 6.3).

5.1.1.1 Bottles. Each bottle shall be packaged in a snug-fitting, corrugated fiberboard box conforming to style RSC (end-opening), type CF, variety optional, class domestic, of PPP-B-636.

5.1.2 Level C.

5.1.2.1 Bottles. Bottles shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice, when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Bottles of one type and size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V3c or V3s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat steel strapping or tape banding shall be in accordance with the appendix of the container specification. The weight of content of each shipping container shall not exceed 65 pounds.

5.2.2 Level B. Unless otherwise specified (see 5.2.2.1), bottles of one type, grade, class, style and size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety optional, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of the container specification. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2.1 When specified (see 6.2), weather-resistant fiberboard specified in 5.2.1 shall be used for the shipping container.

5.2.3 Level C. Bottles, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior boxes and shipping containers shall be marked in accordance with FED-STD-123.

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5.3.2 Military requirements. In addition to any special marking required by the contract or order, the interior boxes and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The vacuum bottles covered by this specification are intended to be used to retain contents at or near the original temperature for a pre-determined length of time.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type and size required (1.2).
c. When preproduction sample is not required (see 3.1).
d. Length of drinking tube required (see 3.4.6).
e. Length of mouth feed tube, 4" or 7" (see 3.4.6.1).
f. If an exterior vent tube is required, and if so, the length required (see 3.4.7).
g. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
h. When weather-resistant fiberboard is required for level B pack (see 5.2.2.1).

6.3 Packaging. The method of packaging referred to in 5.1.1 is similar to Method III of MIL-P-116.

6.4 Civil agency procurement. When Level B packing is required for civil agency procurements, the Level A requirements of 5.1.1 shall apply.

Custodian:
Air Force - 84

Preparing activity:
Air Force - 84

Reviewers:
Air Force - 85, 03

Project number:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Bottle Assembly, Insulated, General Specification For
MIL-B-25337B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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